The Fourier Transform

Every signal is a chord · fill-in notes

woven on Loom

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THE ONE IDEA, AND HOW TO READ THESE NOTES

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> A signal is a sum of **pure tones** $e^{2\pi i\xi x}$, and the Fourier transform $\hat{f}(\xi)$ reads off how much of each is present. The single idea of these notes: **the Fourier basis is the eigenbasis of shift**. Because every translation-invariant operation is diagonal in that basis, the hardest things in one domain (convolution, differentiation, solving a PDE) become the easiest things in the other (just multiplication).

The master dictionary: in space ↔ in frequency (fill it in)

operation on $f(x)$	what it does to $\hat{f}(\xi)$
shift $f(x-a)$	multiply by a phase e________ (§2)
convolution $(f * g)(x)$	ordinary product $\hat{f}(\xi)$ ______ (the headline)
derivative $f'(x)$	multiply by ________
scaling $f(ax)$	stretch/shrink $\frac{1}{\lvert a\rvert}\hat{f}(\xi/a)$
energy $\int \lvert f\rvert^2$	equal energy $\int \lvert\hat{f}\rvert^2$ (Parseval, §3)
narrow in x	____________ wide in ξ (uncertainty, §3)

How to read (passive *and* active)

- **Read** the strands and the stated theorems.
- **Fill** the blanks ______, the [fill in: missing step] proof gaps, and the *Your turn* boxes. Any analysis text is the answer key.
- Bump each `\warmth{0}` to 0–5 once you can rebuild it cold.

CONTENTS

1 THE FOURIER BASIS

□□□□□ A signal is a vector; the pure tones are an orthonormal basis; the transform is the change of coordinates.

Think of a signal f as a vector in an (infinite-dimensional) inner-product space. The **pure tones** $e_\xi(x) = e^{2\pi i \xi x}$ are a basis of that space, one vector per frequency ξ. The Fourier transform is nothing but reading off coordinates in this basis: each $\widehat{f}(\xi)$ is the inner product of f with the tone e_ξ, i.e. ``how much of frequency ξ is in f''.

The transform

Definition 1.1 *(Fourier transform).* For $f : \mathbb{R} \to \mathbb{C}$ (integrable enough),

$$\widehat{f}(\xi) \;=\; \mathscr{F} f(\xi) \;=\; \int_{-\infty}^{\infty} f(x)\, e^{\underline{\qquad\qquad}}\, \mathrm{d}x,$$

the coordinate of f along the tone e_ξ. The finite version is the **DFT**: $\hat{x}_k = \sum_{n=0}^{N-1} x_n\, e^{-2\pi i k n/N}$.

Three transforms to know cold

Your turn. Match each signal to its transform (these recur everywhere):

- a single tone $f(x) = e^{2\pi i \xi_0 x} \longmapsto$ a spike at _________;
- the box $\mathbf{1}_{[-1/2,\,1/2]}(x) \longmapsto \widehat{f}(\xi) =$ ___________ (the sinc);
- the Gaussian $e^{-\pi x^2} \longmapsto$ ___________ (itself!).

Two morals already visible. First, a perfectly localized signal in x (the spike/tone) is perfectly spread in ξ, and vice versa. Second, the transform is a **change of orthonormal basis**, so it loses no information and preserves lengths; that is Parseval's theorem, coming in §2--3.

? Why complex exponentials and not $\sin, \cos$? Because e_ξ are the *eigenfunctions of shift and of* d/dx: shifting or differentiating only multiplies them by a scalar. That is the entire reason Fourier diagonalizes those operations.

? Answers: $\delta(\xi - \xi_0)$; $\mathrm{sinc}(\xi) = \sin(\pi\xi)/(\pi\xi)$; $e^{-\pi\xi^2}$. The Gaussian is the transform's fixed point, and narrow box ↔ wide sinc is uncertainty in miniature (§3).

Conventions vary: some put 2π in front, some in the exponent. We keep it in the exponent, which makes the convolution theorem and Parseval cleanest (no stray constants). Always check a book's convention before borrowing a formula.

2 THE DICTIONARY

□□□□□ Hard in one domain, easy in the other. The headline: convolution becomes multiplication.

The transform earns its fame through a **dictionary**: an awkward operation on f turns into a simple one on $\widehat{f}$. The crown jewel is the **convolution theorem** – smoothing, filtering, and blurring are all convolutions, and in frequency they are just pointwise products.

The translation table

on f	on $\widehat{f}$	name / use
$f(x-a)$	$e^{-2\pi i a\xi}\,\widehat{f}(\xi)$	shift → phase
$e^{2\pi i a x}f(x)$	$\widehat{f}(\xi-a)$	modulation → shift
$(f*g)(x)$	$\widehat{f}(\xi)\,\hat{g}(\xi)$	**convolution → product**
$f(x)\,g(x)$	$(\widehat{f}*\hat{g})(\xi)$	product → convolution
$f'(x)$	$2\pi i\,\xi\,\widehat{f}(\xi)$	derivative → multiply by ξ
$\overline{f(-x)}$	$\overline{\widehat{f}(\xi)}$	symmetry / realness

The two theorems that power the table

Theorem 2.1 *(convolution theorem).* With $(f*g)(x)=\int f(y)\,g(x-y)\,dy$, one has $\widehat{f*g}=\widehat{f}\cdot\hat{g}$.

Proof. Write the transform of $f*g$ as a double integral and substitute $u=x-y$:

$$\widehat{f*g}(\xi)=\iint f(y)g(x-y)e^{-2\pi i\xi x}\,dy\,dx.$$

[fill in: split $e^{-2\pi i\xi x}=e^{-2\pi i\xi y}e^{-2\pi i\xi(x-y)}$ and factor the double integral into $\widehat{f}(\xi)\,\hat{g}(\xi)$]

Theorem 2.2 *(Parseval / Plancherel).* The transform preserves inner products and energy: $\langle f,g\rangle=\langle\widehat{f},\hat{g}\rangle$, and in particular $\|f\|_2=\|\widehat{f}\|_2$.

Your turn. Use the dictionary, not integrals.

- To **filter** a signal (blur it with a kernel g), you convolve $f*g$. In frequency that is just $\widehat{f}\cdot\hat{g}$, so a filter is a ________________ of the spectrum.
- Solving $f''=h$: transforming turns it into $(2\pi i\xi)^2\widehat{f}=\hat{h}$, so $\widehat{f}(\xi)=$ ______________. A differential equation became *division.*

? A filter *reshapes* (multiplies) the spectrum; and $\widehat{f}=\hat{h}/(-4\pi^2\xi^2)$. Calculus → algebra is the whole point.

The discrete mirror: convolution of length-N vectors is diagonalized by the DFT, i.e. every **circulant matrix** is diagonalized by the Fourier matrix, with the transforms as its eigenvalues. Shift-invariance = circulant = diagonal-in-Fourier; one fact, three faces.

3 INVERSION, AND A HARD LIMIT

□□□□□ Orthogonality gives the signal back. And no signal is sharp in both domains at once.

Two facts anchor the theory. The transform is *invertible* (the pure tones are orthonormal, so coordinates determine the vector), and it obeys a *trade-off* (a signal and its transform cannot both be concentrated). The first is why Fourier is useful; the second is why signal processing is interesting.

Getting the signal back

Theorem 3.1 *(inversion).* If $\widehat{f}$ is nice enough, then

$$f(x) = \int_{-\infty}^{\infty} \widehat{f}(\xi)\, e^{2\pi i \xi x}\, \mathrm{d}\xi.$$

That is: f is the superposition of its tones, each weighted by its coordinate $\widehat{f}(\xi)$.

Proof. Skeleton: the tones are orthonormal in the sense $\int e^{2\pi i\xi x}\, \overline{e^{2\pi i\xi' x}}\, \mathrm{d}x = \delta(\xi - \xi')$. [fill in: substitute $\widehat{f}(\xi) = \int f(y) e^{-2\pi i \xi y}\, \mathrm{d}y$ into the right side and collapse the ξ-integral with the orthogonality relation]

? So $\mathcal{F}$ is a unitary change of basis on $L^2(\mathbb{R})$: it preserves lengths (Parseval) and is undone by $\mathcal{F}^{-1}$, which is the same formula with a + sign in the exponent.

The uncertainty principle

Theorem 3.2 *(Heisenberg uncertainty).* For a unit-energy signal, let $(\Delta x)^2$ and $(\Delta\xi)^2$ be the variances of $|f|^2$ and $|\widehat{f}|^2$. Then

$$\Delta x \cdot \Delta\xi \ \geq\ \frac{1}{4\pi},$$

with **equality exactly for Gaussians**. You cannot be sharply localized in space and in frequency at once.

This is not a defect of the transform; it is a law. A pure tone (one frequency) is spread over all space; a spike (one instant) needs all frequencies. The Gaussian sits exactly at the boundary, equally (and minimally) spread in both, which is why it is the transform's fixed point and the shape of every ``optimal'' window, wave packet, and kernel.

Your turn. Scale a Gaussian: let $f_a(x) = e^{-\pi a x^2}$.

- Its transform is $\widehat{f_a}(\xi) = \frac{1}{\sqrt{a}} e^{-\pi\xi^2/a}$. As $a \to \infty$ the signal gets ____________ in x and ____________ in ξ.
- So the product of the two widths stays ______________ — the uncertainty bound is tight here.

? Narrower in x, wider in ξ (and vice versa); the width-product is constant. Squeeze one domain, the other balloons.

The same inequality, read in quantum mechanics with x = position and ξ = momentum, is Heisenberg's. Position and momentum are a Fourier pair; the ``principle'' is a theorem about $\mathcal{F}$.

4 FACES OF THE TRANSFORM

□□□□□ Fast computation, sampling, filtering, PDEs: one dictionary, cashed out.

> Because Fourier diagonalizes every shift-invariant operation, a long list of classical problems collapse to ``transform, multiply, transform back''. Here is the keyring.

The keyring

face	the idea
the **FFT**	compute the length-N DFT in $O(N \log N)$, not $O(N^2)$, by splitting even/odd indices and recursing
sampling (Nyquist)	a signal band-limited to $\lvert\xi\rvert \le B$ is fully determined by samples at rate $\ge 2B$
filtering	multiply $\widehat{f}$ by a mask = convolve f with the kernel; low-pass, high-pass, denoise
solving **PDEs**	the heat equation $u_t = u_{xx}$ becomes $\hat{u}_t = -(2\pi\xi)^2\hat{u}$: an ODE *per frequency*
circulant matrices	all diagonalized by the Fourier matrix; eigenvalues are the ˆs

Theorem 4.1 *(the FFT, in one line).* Splitting a length-N DFT into its even- and odd-indexed halves gives $\hat{x}_k = \widehat{(\text{even})}_k + e^{-2\pi i k/N}\,\widehat{(\text{odd})}_k$, two DFTs of size $N/2$ plus $O(N)$ glue. Recursing costs $T(N) = 2T(N/2) + O(N) = O(N \log N)$.

Your turn. The heat equation, diagonalized. Transforming $u_t = u_{xx}$ in x turns it into a simple ODE in t for each frequency:

$$\hat{u}_t(\xi,t) = -(2\pi\xi)^2\,\hat{u}(\xi,t) \quad\Longrightarrow\quad \hat{u}(\xi,t) = \underline{\hspace{6cm}}\,.$$

High frequencies therefore decay __________ than low ones. (That is exactly why heat smooths.)

? $\hat{u}(\xi,t) = \hat{u}(\xi,0)\,e^{-(2\pi\xi)^2 t}$; high frequencies decay *faster* (the ξ^2 in the exponent). Smoothing = killing high frequencies.

The one idea, recovered

> Every face above is the same sentence: **Fourier diagonalizes shift**. Convolution, differentiation, circulant matrices, and constant-coefficient PDEs are all built from translation, so all of them become multiplication once you change to the frequency basis. Read the master table again; you can now fill, and explain, every row.

Where to go next: replace $\mathbb{R}$ by a finite group and you get the DFT and group representations; demand localization in *both* domains and you get wavelets; take the group to be $\{0,1\}^n$ and you get the **Fourier analysis of Boolean**